MIL-E-1 1334F(NAVY)
4 February 1974
SUPERSEDING
MIL-E-1/1334E(NAVY)
28 April 1971

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NEGATIVE GRID (MICROWAVE)

TYPE 7296

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, metal-ceramic
See figure 1
Mounting position: Any
Weight: 0.3 ounce nominal

ABSOLUTE RATINGS:

Parameter:	F1	Ef	Eb	Ec	Ehk	Rk	Rg	
Unit:	MHz	V	Vdc	Vdc	Vdc	Ohms	Meg	
Maximum:	400	6.6	330	---	+50	---	0.1	(Fixed bias)
							0.18	(Cathode bias)
Minimum:	---	6.0	---	-50	-50	---	---	
TEST CONDITIONS:	---	6.3	200	0	---	68	---	

ABSOLUTE RATINGS:

Parameter:	Ik	ik	Ic	Pp	TE	(F) Barometric pressure, reduced
Unit:	mAdc	ma	mAdc	W	°C	mmHg
Maximum:	30	120	10	5.5	Note 1	8
Minimum:	---	---	---	---	---	---
TEST CONDITIONS:	---	---	---	---	---	---

GENERAL:

Qualification - Required

(F) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1301	Heater current	-		0.65	II	If	370	430	mA
1336	Heater-cathode leakage	-		0.65	II	Ihk	---	20	μAdc
1266	Total grid current	-	Ecc = -1.0 Vdc; Rg = 0.18 Meg; Rk = 68 ohms	0.65	II	Ic	---	-0.5	μAdc
1306	Transconductance (1)	-	Eb = 100 Vdc; Rk = 0	0.65	II	Sm	13,000	---	μmhos
1231	Pulsing emission	2	Rk = 1.0 ohm	0.65	II	Ik	320	---	ma
(F) 1201	Short and discontinuity detection	-		0.65	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes	-		---	---	R	100	---	Meg
1306	Transconductance (2)	-	Ef = 6.0 V; Eb = 100 Vdc; Rk = 0	---	---	ΔSm Ef	---	20	%
1256	Electrode current (anode)	-		---	---	Ib	10	24	mAdc
1306	Transconductance (3)	-		---	---	Sm	13,000	20,000	μmhos
1316	Amplification factor	-		---	---	Mu	65	115	---
1261	Electrode voltage (grid)	-	Rk = 0; Ec/Ib = 10 μAdc	---	---	Ec	---	-9.5	Vdc
1266	Grid emission	3	Ef = 7.0 V; Ecc = -15 Vdc; Rk = 0; Rg = 0.18 Meg	---	---	Ic	0	-2.0	μAdc
(F) 2210	Grid recovery	5	Ebb = 250 Vdc; Ec/Ib = 10 mAdc; Rp = 0.01 Meg	---	---	ΔIb Δib	--- ---	1.0 2.0	mAdc ma
1256	Power oscillation	-	F = 400 MHz; Ebb = 300 Vdc; Ib = 20 mAdc; Rg = 1,400 ohms; Ic = 6.0 to 9.0 mAdc	---	---	Po	1.6	---	W (useful)
1331	Direct-interelectrode capacitance	-		---	---	Cgp Cin Cout Chk	1.9 3.7 0.05 2.1	2.5 6.3 0.1 3.5	pF pF pF pF
---	Envelope strain	10		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3								
1506	Heater-cycling life	11	Ef = 7.0 V; Ehk = +70 Vdc; Rk = 0; Ec = Eb = 0	---	---	---	---	---	---
---	Heater-cycling life-test end point:								
1336	Heater-cathode leakage	-		---	---	Ihk	---	40	μAdc
(F) 1516	Stability life	12	Ebb = 250 Vdc; Rk = 200 ohms; Eg = 65 Vac; Ehk = -70 Vdc; Rb = 400 ohms	2.5	Code F	---	---	---	---
---	Stability life-test end point: (2 and 20 hours)								
---	Change in transconductance (1) of individual tubes	-		---	---	ΔSm t	---	20	%
(F) 1501	Intermittent life	-	Group A						
---	Operation (1)	-	Stability life-test conditions: t = 1,000 hours	---	---	---	---	---	---
---	Operation (2)	-	Ebb = 250 Vdc; Ec = -20 Vdc; Ehk = -70 Vdc; ec = +15 (driver side of Rg); Rg/g = 50 ohms; Rb = 60 ohms; tp = 10 ±1 μs; Du = 0.01 ±0.1%; tr = 1.5 μs (max); tf = 2.5 μs (max); t = 1,000 hours	---	---	---	---	---	---
---	Intermittent life-test end points:								
---	Operation (1)								
1301	Heater current	-		---	---	If	370	440	mA
1306	Transconductance (1)	-		---	---	Sm	10,500	---	μmhos
1306	Transconductance (2)	-		---	---	ΔSm Ef	---	25	%
1236	Power oscillation	-		---	---	Po	1.2	---	W (useful)
1336	Heater-cathode leakage	-		---	---	Ihk	---	20	μAdc
---	Operation (2)								
1301	Heater current	-		---	---	If	370	440	mA
1231	Pulsing emission	-		---	---	ik	260	---	ma
1336	Heater-cathode leakage	-		---	---	Ihk	---	20	μAdc

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
(F)	Periodic-check tests								
1002	Barometric pressure, reduced	4	Pressure = 8 ±1 mmHg; voltage = 300 Vac	---	---	---	---	---	---
1031	Sweep-frequency vibration	6	Ebb = 200 Vdc; Rk = 68 ohms; Rp = 2,000 ohms, 10 G	---	---	Ep	---	50	mVac
1031	Low-frequency vibration	7	Ebb = 200 Vdc; Rk = 68 ohms; Rp = 2,000 ohms; 15 G; F = 40 Hz	---	---	Ep	---	50	mVac
1041	Shock	8	Hammer angle = 42°; Eb = 200 Vdc; Rk = 68 ohms; Ehk = 100 Vdc; Rg = 0.1 Meg	---	---	---	---	---	---
1031	Sweep-frequency vibration fatigue	9	Eb = Ec = 0; Ef = 6.3 V; sweep-frequency vibration	---	---	---	---	---	---
---	Shock and sweep-frequency vibration fatigue test end points:								
1031	Low-frequency vibration	-		---	---	Ep	---	30	mVac
1336	Heater-cathode leakage	-		---	---	Ihk	---	30	μAdc
1306	Transconductance (1)	-		---	---	Sm	11,000	---	μmhos
1301	Heater current	-		---	---	If	370	440	mA

NOTES:

1. Maximum envelope temperature:

   Anode dissipation not over 3.3 watts - - - +300° C
   Anode dissipation not over 5.5 watts - - - +250° C

2. Ec = -20 Vdc. Grid is driven with pulse circuit as follows: ec = +12 v; prr = 1,000; tp = 10 ±1 μs; tr = 1 μs (max), tf = 1 μs (max). Read peak cathode current.

3. Prior to this test, tubes shall be preheated a minimum of 5 minutes at the conditions indicated below. Test within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

Ef V	Ecc Vdc	Eb Vdc	Rk ohms	Rg Meg	Rp Meg
7.0	0	300	50	0.18	0.003

4. The tube shall be tested as specified in method 1002, with the exception that the specified voltage shall be applied between the anode and grid.

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NOTES: - Continued.

(F) 5. Test shall be performed as follows:

(a) With TUT operating under specified conditions, adjust Ec for Ib = 10 mAdc.
(b) Apply 3-volt pulse, 20 μs duration, 60 pps to the grid.
(c) With application of pulse, measure undershoot (Δib) and change in average current (ΔIb).

6. Sweep-frequency vibration. Test shall be performed as follows:

(a) The frequency shall be increased from 100 to 2,000 Hz with approximately logarithmic progression in 3 ±1 minutes. The return sweep (2,000 to 100 Hz) is not required.
(b) The tube shall be vibrated with simple harmonic motion in each of two planes; first, parallel to the cylindrical axis; second, perpendicular to the cylindrical axis and parallel to a line through the major axis of a terminal lug. At all frequencies from 100 to 2,000 Hz, the total harmonic distortion of the acceleration waveform shall be less than 5 percent.
(c) The peak acceleration shall be maintained at 10 ±1.0 G throughout the test.
(d) The value of the alternating voltage, Ep, produced across the resistor, Rp, as a result of the vibration shall be measured with a suitable device having a response to the rms value of the voltage to within ±0.5 dB of the response at 400 Hz for the frequency range of 100 to 3,000 Hz, and having a bandpass filter with an attenuation rate of 24 dB per octave below the low-frequency cut-off point of 50 Hz and above the high-frequency cut-off point of 5,000 Hz. The meter shall have a dynamic response characteristic equivalent to or faster than a VU meter (operated in accordance with Standard ASA No. C 16.5 - 1954).

7. Low-frequency vibration. Test shall be performed as follows: The tube shall be vibrated with simple harmonic motion in each of two planes; first, parallel to the cylindrical axis; second, perpendicular to the cylindrical axis and perpendicular to major axis of terminal lug.

8. The tubes shall be rigidly mounted by T-bolt on the shock table and voltages shall be applied to the terminal lugs through flexible leads. A grid resistor of 0.1 megohm shall be added; however, this resistor shall not be used when a thyratron-type short indicator is used.

9. Sweep-frequency vibration fatigue. Test shall be performed as specified in method 1031, except that the tubes shall be vibrated for a total of 6 hours, that is, 3 hours in each of two directions; first, parallel to the cylindrical axis; second, perpendicular to the cylindrical axis and parallel to a line through the major axis of a terminal lug.

10. Tubes shall be tested as specified in method 2126, except that they shall first be immersed in water at not more than 5°C for 15 seconds and immediately thereafter subjected to the standard temperature cycle specified in method 2126.

(F) 11. The heater-cycling life-test sample shall consist of 15 tubes and no tube failures shall be permitted. In the event of rejection of the first sample due to failure of 1 tube, a second sample of 15 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples. The total tube failures from the combined first and second samples shall not exceed one.

12. Eg = 65 Vac, 60 Hz, applied to grid through a series resistor of 8,000 ohms bypassed with a 1.0 μF capacitor.

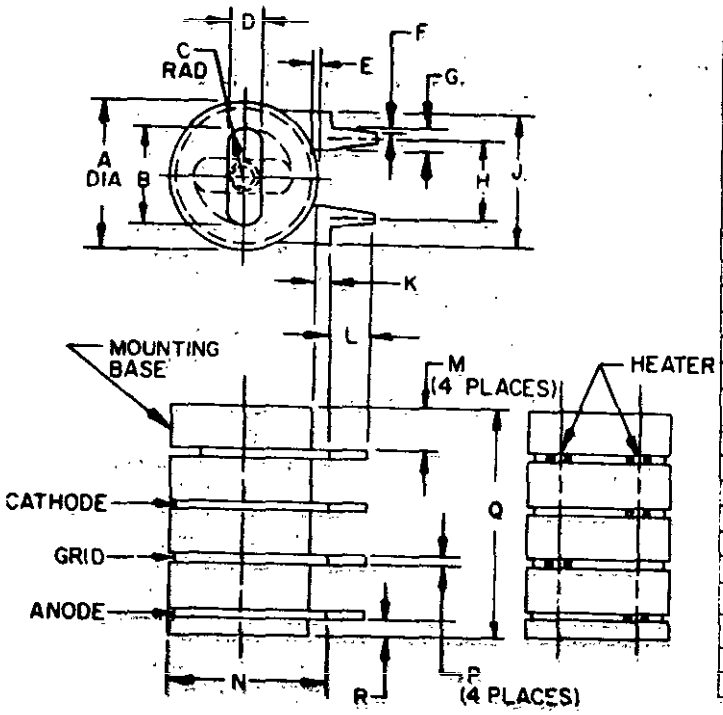
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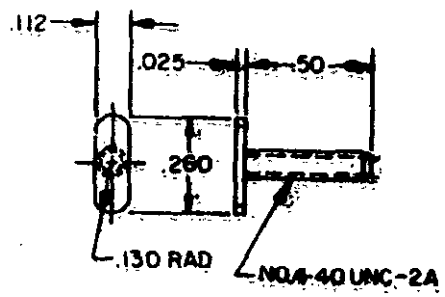
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Ltr	Dimensions in inches with metric equivalents (mm), in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1 (See note b)		
A	.535(13.59)	.565(14.35)
B	.290 (7.37)	.310 (7.87)
D	.115 (2.92)	.145 (3.68)
G	.080 (2.03)	.086 (2.18)
N	.595(15.11)	.635(16.13)
P	.025 (.64)	.031 (.79)
Q	.860(21.84)	.890(22.61)
Quality conformance inspection, part 2		
C	.140 (3.56)	.160 (4.06)
H	.290 (7.37)	.310 (7.87)
J	.490(12.45)	.510(12.95)
L	.145 (3.68)	.155 (3.94)
M	.170 (4.32)	.180 (4.57)
R	.070 (1.78)	.080 (2.03)
Reference dimensions		
E	.025 (.64)	
F	.006 (.15)	
K	.065 (1.65)	

INCHES	MM
.025	.64
.112	2.84
.130	3.30
.260	6.60
.50	12.70



MOUNTING BOLT

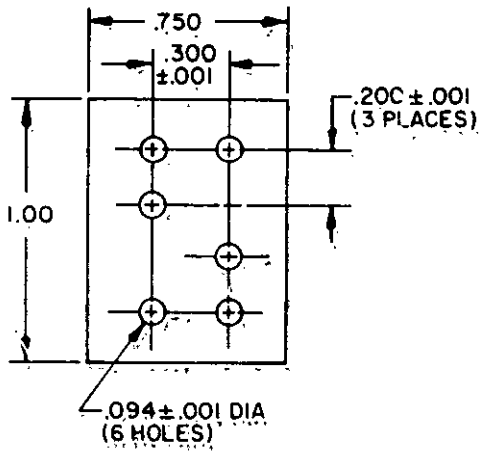
NOTES:

a. All terminals shall be so alined that they are capable of being inserted into a gage .125(3.18 mm) thick having 6 holes .094(2.39 mm) ± .001(.03 mm) in diameter as shown in figure 2.

(F) b. The AQL for dimensions listed under quality conformance inspection, part 1, shall be 1.0 percent, inspection level II.

FIGURE 1. Outline drawing of electron tube type 7295.

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INCHES	MM
.001	.03
.094	2.39
.200	5.08
.300	7.62
.750	19.05
1.00	25.40

FIGURE 2. Terminal-position gage.

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. *GIVE PARAGRAPH NUMBER AND WORDING.*

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY (Printed or typed name and activity - Optional)

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